

The empirical recurrence for OEIS A299882, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A299882 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 71$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 69$. Its last coefficient is $c_{66} = 1620 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299882 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 3, 4, 5, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 16, 28, 247, 677, 3597, 21500, 101350, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Jan 27 2022 20:44:48, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 71$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 142$ exact terms from $n = 3$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	5	c_{18}	−2512818	c_{35}	−1070145902	c_{52}	−308421522
c_2	5	c_{19}	12317022	c_{36}	−676188992	c_{53}	−537274407
c_3	54	c_{20}	932602	c_{37}	753792877	c_{54}	−264115311
c_4	−362	c_{21}	9899316	c_{38}	899481560	c_{55}	−37196867
c_5	−262	c_{22}	−28896658	c_{39}	295208990	c_{56}	137138452
c_6	−670	c_{23}	−5780050	c_{40}	−545390153	c_{57}	148989102
c_7	9160	c_{24}	−9320765	c_{41}	−776685583	c_{58}	46390371
c_8	3970	c_{25}	42980839	c_{42}	−268607936	c_{59}	−2450298
c_9	2134	c_{26}	32409415	c_{43}	654332553	c_{60}	−21948484
c_{10}	−115398	c_{27}	−23222096	c_{44}	937657401	c_{61}	−13638204
c_{11}	−27463	c_{28}	25204056	c_{45}	141113016	c_{62}	−3925649
c_{12}	−8296	c_{29}	−181048325	c_{46}	−919348222	c_{63}	−1109754
c_{13}	829111	c_{30}	−136405563	c_{47}	−1427484200	c_{64}	−115113
c_{14}	129382	c_{31}	−62064684	c_{48}	−149296536	c_{65}	17595
c_{15}	285248	c_{32}	651413680	c_{49}	939689754	c_{66}	1620
c_{16}	−3815637	c_{33}	629069477	c_{50}	1010683967		
c_{17}	−485981	c_{34}	−361435522	c_{51}	377067359		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 69$, and it is the recurrence OEIS A299882 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = -1620$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1620.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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